CN Practical 4: Create a Small LAN

Aim: To create a small Local Area Network and understand the different network control devices used in a LAN.

Theory:

A Local Area Network (LAN) is a computer network that connects devices within a limited geographical area, such as a computer laboratory, office, home or college building. A LAN allows computers to communicate, share files, access printers and use a common internet connection. LANs are generally faster and easier to manage than networks covering large geographical areas.

A switch is one of the main devices used in a wired LAN. It forwards Ethernet frames to the appropriate destination device using MAC addresses. A router connects the LAN to other networks and can provide internet access. Network Interface Cards (NICs), Ethernet cables and IP addresses are also required for communication. Devices in the same IP subnet can generally communicate directly without requiring a router.

Materials required:

- Two or more computers
- Ethernet cables
- Network switch
- Router, if internet access is required
- Windows operating system

Example IP configuration:

Device	IP address	Subnet mask
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Procedure:

1. Place the computers and network switch in the required positions.
2. Connect each computer to the switch using an Ethernet cable.
3. Check that the Ethernet interfaces are enabled and the physical links are active.
4. Open the network settings on each computer.
5. Assign unique IP addresses from the same subnet.
6. Configure the subnet mask as 255.255.255.0.
7. Set the default gateway if communication with other networks is required.
8. Open Command Prompt on PC1.
9. Execute ipconfig to verify its IP configuration.
10. Execute ping 192.168.1.3 to test connectivity with PC2.
11. Repeat the connectivity test between other computers.

Observation:

The computers were connected through a central switch and configured with unique IP addresses in the same subnet. The ipconfig command displayed the assigned network settings, and the ping command was used to verify communication. Successful replies indicated that the computers could exchange packets over the LAN.

Conclusion:

A small Local Area Network was created using computers, Ethernet cables and a network switch. The computers were assigned appropriate IP addresses and tested using network diagnostic commands. The practical demonstrated the basic operation and resource-sharing capabilities of a LAN.